Written Exposure Therapy for Transdiagnostic Anxiety Symptoms: Treatment Efficacy and Mechanisms of Change

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Abstract

This study investigates mechanisms underlying Transdiagnostic Written Exposure Therapy (WET), a novel, scalable online intervention for pathological anxiety symptoms. In a randomized controlled trial with 89 participants, WET produced significant anxiety reduction that continued through two-month follow-up (SMD=0.85). We systematically evaluated multiple potential mechanisms across cognitive, emotional, and behavioral domains. From this broad assessment, two candidate mechanisms emerged as both significantly affected by treatment and correlated with symptom improvement: enhanced distress tolerance (r = -0.35) and decreased anxious responses to core threats (r = 0.39). This suggests exposure therapy might operate through dual pathways: improving distress tolerance and reducing anxious responses. Interestingly, a neutral imagery control condition showed immediate but non-sustained anxiety reduction and actually decreased distress tolerance, revealing distinct therapeutic trajectories between neutral imagery-based versus exposure-based interventions. This research advances understanding of mechanisms of exposure while highlighting specific targets for enhancing interventions.

Keywords: Core threats, Transdiagnostic anxiety, Written Exposure Therapy

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Written interventions have long been part of the psychological toolbox (Ruini & Mortara, 2022), both as an aid and as a standalone technique (Riordan, 1996; Smyth et al., 2008). One such method, expressive writing, is a therapeutic technique in which individuals freely write about emotional or traumatic experiences (Pennebaker, 1997). Evidence indicates that expressive writing has a consistent, but limited effect, which can be more effective in focused contexts (e.g., Frattaroli, 2006).

Written Exposure Therapy (WET) is a structured, brief intervention designed to treat individuals with PTSD through expressive writing (Sloan & Marx, 2019). Originally developed as a variation of imaginal exposure, WET was created to offer a streamlined, accessible alternative to more traditional therapies. WET yields strong effects for PTSD, comparable to established treatments like Cognitive Processing Therapy and Prolonged Exposure (Sloan et al., 2023; Thompson-Hollands et al., 2019). While WET has traditionally been applied specifically to PTSD, it has been extended to generalized anxiety disorder by focusing on future-oriented feared outcomes (Goldman et al., 2007; Ovanessian et al., 2019) including in an online format (Roch-Gagné & Talbot, 2019), indicating its potential for broader, transdiagnostic applications.

We introduce transdiagnostic WET, a future oriented version of WET tailored for individuals with transdiagnostic anxiety symptoms. Since WET integrates elements of both imaginal exposure (IE) and expressive writing, we address the foundations of each approach to fine-tune transdiagnostic WET. Two key insights drive the modifications that have been made: exposure needs to be a) focused and b) immersive.

Directive instructions yield better outcomes in expressive writing (Frattaroli, 2006; Reinhold et al., 2018). Indeed, WET for PTSD focuses on the most distressing traumatic

event that the client has (Sloan & Marx, 2019). Furthermore, WET has been found to be more effective when focusing on a single fear rather than alternating between different fears, in both GAD and PTSD (Fracalanza et al., 2014; Sloan et al., 2005). In PTSD, WET focuses on the central source of anxiety - the traumatic event. We propose that transdiagnostic WET should focus on core threats: the ultimate feared consequence of one's fear or anxiety (Huppert & Zlotnick, 2012; Zlotnick & Huppert, 2025). Clinicians suggest that focusing on core threats improves efficacy, enhances generalizability, and reduces relapse (Foa & Kozak, 1986; Gillihan et al., 2012; Murray et al., 2016). We have developed an online questionnaire to reliably identify core threats (Zlotnick & Huppert, 2026). Thus, to ensure that the intervention is accurately focused, we help participants identify their core threats and then instruct them to focus on these idiosyncratic core threats throughout their writing.

A second set of accommodations to WET is creating an immersive script. Building on the work of Lang (1977), emotional processing theory contends that emotional activation is necessary for emotional processing to occur (Foa et al., 2006; Foa & Kozak, 1986). Indeed, IE protocols typically involve conjuring a vivid, detailed image of one's worst case scenario from a first person perspective (e.g., Abramowitz, 2006; Foa et al., 2012). Furthermore, focusing on the emotional aspects in writing is more beneficial than focusing on cognitive or control conditions (Sloan et al., 2007). Moreover, individuals undergoing WET show signs of physiological distress, supporting the idea that emotional activation is necessary for successful treatment (Wisco et al., 2016). However, the WET protocol for PTSD emphasizes recounting traumatic experiences from a distancing perspective (e.g., "as you look back upon it now") rather than an immersed perspective (e.g., "as if it were happening right now;" Sloan & Marx, 2019). This approach is based on the idea that a distanced perspective facilitates "working through" difficult life experiences (Kross & Ayduk, 2017). We suggest that this approach is appropriate for PTSD, where a major aspect of the intervention is reconstructing the traumatic narrative. Indeed, the WET protocol explicitly states this goal: "By repeatedly recounting the event, you will be able to correct for the splintered fashion in

which the memory may have been stored" (Sloan & Marx, 2019, p. 53). We contend that in future-oriented WET for a non-traumatized population increased emotional activation is likely necessary for optimal outcomes. Thus, participants are directed to immerse themselves in the writing by using first person, providing detailed sensory and emotional descriptions, and periodically closing their eyes and calling upon their imaginations.

The first goal of this study was to develop and validate transdiagnostic WET. Our second goal was to explore several theory-guided mechanisms of change (cf. Rousseau & Belleville, 2018), as understanding mechanisms can inform more precise interventions (Niv et al., 2021). Following Kazdin's framework (Kazdin, 2007), this study intentionally adopted an exploratory approach to map multiple processes spanning emotional, cognitive, and behavioral domains.

Specifically, we examined:

1. **Fear reduction:** Exposure may reduce perceived threat by updating how feared situations are interpreted. This shift can occur through various learning processes, including habituation, extinction, or inhibitory learning (Craske et al., 2008; Foa & Kozak, 1986).
2. **Distress tolerance:** The ability to withstand negative emotional states may be more critical to exposure therapy success than fear reduction itself (Craske et al., 2008; Hayes et al., 1999).
3. **Thought-action fusion:** The belief that thoughts directly influence reality is associated with anxiety disorders and OCD (Rassin et al., 1999; Thompson-Hollands et al., 2013). We hypothesized that experiencing negative imagery without consequences would reduce thought fusion, improving outcomes.
4. **Negative beliefs about imagery and memory:** Beliefs that engaging with distressing content is dangerous can sustain avoidance. WET may challenge these beliefs by promoting safe emotional engagement with feared scenarios, thereby

facilitating cognitive restructuring.

5. **Reactions to future imagery:** Personally feared future scenarios can evoke distress, intrusiveness, and avoidance distinct from general anxiety or metacognitive beliefs. Assessing these reactions clarifies whether WET changes emotional responses to feared future events themselves—such as distress when imagining failure, loss, or rejection—rather than only broader cognitive appraisals.
6. **Self-compassion:** A capacity for non-judgmental acceptance of one's emotional experience that can counter self-criticism and avoidance, promoting approach toward feared emotions.

Together, these mechanisms represent distinct but complementary processes that may underlie the effects of imaginal exposure.

This study aims to extend WET in three ways: a) applying it to a transdiagnostic population; b) exploring its utility for coping with future threats rather than past events; and c) assessing its effectiveness as a self-administered, online intervention. To date, WET has not been widely tested within this type of sample or setting. We hypothesized that WET will yield greater reductions in anxiety symptoms and improvements in well-being compared to both active control (written distraction imagery) and waitlist control conditions.

An additional aim is to investigate the potential mechanisms underlying WET, enhancing our understanding of how it facilitates therapeutic change. More insight into mechanisms could help fine tune theory, optimize interventions and improve outcomes (Huppert et al., 2020; Niv et al., 2021). We measured potential mechanisms that might drive WET's effects, focusing on both threat-specific processes (e.g., fear reduction, reactions to distressing memories) and threat-agnostic processes [e.g., distress tolerance, thought fusion, and negative metacognitions regarding memories or images; Rousseau and Belleville (2018)].

To explore these processes, we examined whether WET changes these processes and,

if so, whether these changes correlate with reductions in anxiety symptoms. We anticipated that changes in mechanisms will align with symptom changes, supporting their role in the intervention's therapeutic process.

Transparency and Openness

Preregistration:

This study was preregistered prior to data collection at doi.org/10.17605/OSF.IO/2CEHR.

Deviations from preregistration:

We departed from the preregistration on the following points; all decisions were made before data collection.

- We added a passive waitlist control group, drawn from a prior study with identical inclusion, exclusion, and procedural criteria on the same recruitment platform (see Sorka et al., in preparation). Because this group was not randomized concurrently with the intervention arms, results involving it should be interpreted as comparative rather than inferential.
- We administered follow-up assessments at one week and two months rather than at a single unspecified follow-up.
- We replaced the Thought Fusion Inventory with the Thought-Action Fusion Scale Revised, and administered it pre- and post-intervention rather than at every writing session.
- We specified the growth model's time component as piecewise rather than as a single linear slope, reflecting the qualitatively different change regimes across the intervention and the two follow-up windows.
- The Imaginal Behavior Approach Test was administered as preregistered but not analyzed in the present report.

Data, materials, code, and online resources:

Data, analysis scripts, participant-facing materials, and methodological documentation are available at https://github.com/eladzlot/transdiagnostic.wet-2025-public and mirrored at the OSF project https://osf.io/dfzgt/. The repository is released under the MIT License. Datasets include only quantitative information to protect participant confidentiality. Participant-facing materials, including the consent text, intervention instructions, and a navigation document with response anchors and citations for every measure used in the study, are provided in `supplementary/materials/`.

Reporting:

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. A power analysis indicated a target of 148 participants. Recruitment was halted before this target was reached under a preregistered Bayesian stopping rule; specifics are given in the Methods.

Ethical approval:

This research was approved by the Ethics Committee of The Hebrew University of Jerusalem. All participants provided informed consent.

Methods

This study was a randomized controlled trial (RCT) in which participants were randomly assigned to one of two groups: the treatment group (WET) or the control group (Neutral Imagery Writing, NIW). Additionally, data from a waitlist control group was incorporated, collected from a prior study with the same structure and inclusion/exclusion criteria (see Sorka et al., in preparation). This provided both a passive control and an active control for comparison. The study employed a double-blind design to ensure objectivity. Study personnel had no direct interaction with participants, and both experimenters and participants were unaware of group assignments until the one-week follow-up. The primary outcome was defined as the change in OASIS scores, with changes in functioning scores and

other proposed mechanisms serving as secondary outcomes. Covariations between changes in OASIS and changes in each potential mechanism were also examined.

Participants

Participants were recruited through the Prolific platform, compensated at £9/hour. Inclusion criteria included moderate pathological anxiety (OASIS score > 4) and impaired daily functioning (at least one WSAS item scored above 2 on a 0–8 scale). Exclusion criteria were severe depression (greater than 14 on the PHQ; Kroenke et al., 2009), elevated post-traumatic symptoms (greater than 6 on the brief PCL-5; Zuromski et al., 2019), or presence of psychotic symptoms (evaluated via items 19 and 20 from the DIAMOND screener; Tolin et al., 2018). Furthermore, participants who had reported head injury or reading and writing difficulties in the Prolific system were excluded. In addition, English as a fluent language was required. Finally, only experienced users on Prolific with an approval rate above 95, and minimum 300 previous submissions were allowed to participate. An ethnically diverse sample was requested from Prolific.

Power analysis indicated that a sample size of 148 participants would provide 0.8 power. Due to financial constraints, we preregistered a Bayesian stopping rule allowing for early termination: participants were recruited in increments of 30, and after each increment we evaluated whether 90% of the posterior distribution for the WET–NIW difference in end-of-treatment OASIS fell within a region of practical equivalence of ±2 points. Recruitment stopped when this criterion was met.

Participants were randomly assigned to the treatment and control groups in a 2:1 ratio, using stratified randomization ($OASIS < or \geq 8$), aiming to ensure equal distribution of anxiety severity between groups. The block randomization process was integrated into the experiment program code.

Measures

The *Overall Anxiety Severity and Impairment Scale* (OASIS; Norman et al., 2006) assesses the frequency, intensity, and functional impairment of anxiety and fear over the past week. Participants rate their anxiety on a 5-point scale from 0 (little or none) to 4 (extreme or all the time). Higher scores indicate more severe anxiety-related impairment, with a cut-off score of 8 recommended for identifying anxiety disorders and a change of four points considered clinically significant (Moore et al., 2015). The OASIS has strong psychometric properties, with an omega coefficient of 0.86 across timepoints in our sample, indicating high reliability.

The *Work and Social Adjustment Scale* (WSAS; Mundt et al., 2002) was used to assess functional impairment. The scale consists of five items, each rated on a 9-point Likert scale (0 = no impairment to 8 = severe impairment), evaluating the extent of difficulty in work, home management, social leisure, private leisure, and close relationships. In our sample, the omega coefficient was 0.80, indicating high reliability.

The *Thought-Action Fusion Scale Revised* (TAF-R; Shafran et al., 1996) measured the degree to which individuals equate thoughts with actions. It includes 19 items rated on a 5-point Likert scale (0 = disagree strongly to 4 = agree strongly), with higher scores indicating a stronger tendency toward thought–action fusion. The scale comprises two subscales: TAF-Moral, which reflects the belief that having unacceptable thoughts is morally equivalent to performing them, and TAF-Likelihood, which captures the belief that thinking certain thoughts increases the likelihood of them occurring. We analyzed these subscales separately. In our sample, the omega coefficients for moral and likelihood TAF were 0.94 and 0.97 respectively, indicating excellent reliability.

The *Brief Core Schema Scale* (BCSS; Fowler et al., 2006) was employed to assess cognitive schemas. It consists of 24 items regarding the self or others (e.g., “I am talented”;

"I am loved"), divided into two subscales each (e.g., self-positive and self-negative). Each item is rated on a 5-point Likert scale (0 = Don't Believe it to 4 = Believe it totally). In our sample, the omega coefficients for each scale were at least 0.87, indicating excellent reliability.

The *Distress Tolerance Scale* (DTS; Simons & Gaher, 2005) assessed participants' ability to tolerate distressing emotions. The DTS consists of 15 items rated on a 5-point Likert scale (1 = strongly agree to 5 = strongly disagree), evaluating various aspects of distress tolerance. Due to a coding mistake, item 6 was missing and item 7 was repeated twice, thus we dropped item 6 completely from all analyses. In our sample, the omega coefficient was 0.93, indicating excellent reliability.

The *Metacognition Questionnaire-30* (MCQ-30; Wells & Cartwright-Hatton, 2004) was employed to evaluate metacognitive beliefs related to anxiety. The MCQ-30 consists of 30 items rated on a Likert scale ranging from 1 (do not agree) to 4 (agree very much). It assesses five factors of metacognition: cognitive confidence, positive beliefs about worry, cognitive self-consciousness, negative beliefs about the uncontrollability of thoughts and danger, and beliefs about the need to control thoughts. In our sample, the omega coefficients were at least 0.89, indicating excellent reliability.

The *Self-Compassion Scale-Short Form* (SCS-SF; Raes et al., 2011) assessed participants' levels of self-compassion. The SCS-SF consists of 12 items rated on a 5-point Likert scale (1 = almost never to 5 = almost always). Higher scores indicate a greater frequency of self-compassionate thoughts and behaviors. In our sample, the omega coefficient was 0.85, indicating excellent reliability.

The *Impact of Future Event Scale* (IFES; Deeprose et al., 2011) assessed the impact of intrusive prospective, personally-relevant imagery. Participants completed 24 items evaluating intrusive pre-experiencing, avoidance, and hyper-arousal, responding to how frequently each statement was true for them over the past seven days. Responses were given

on a 5-point scale (0 = not at all to 4 = frequently). In our sample, the omega coefficient was 0.89, indicating excellent reliability.

The *Brief State Anxiety Measure* (BSAM; Berg et al., 1998) was used to assess participants' state anxiety in response to their core threat. The BSAM consists of six items (relaxed, steady, strained, comfortable, worried, tense) adapted from the State-Trait Anxiety Inventory. These items are designed to provide a brief and efficient measure of state anxiety. Each item is rated on a 7-point scale (1 = Not at all, 7 = Very much so), with higher scores indicating greater levels of anxiety. In this study, participants were asked to imagine their core threat script for a few moments both before and after each writing session. They then completed the BSAM to report their current anxiety levels. The BSAM demonstrated excellent internal consistency in our sample, as indicated by an omega coefficient of 0.96.

The *Treatment Credibility Questionnaire* (Devilly & Borkovec, 2000) was used to evaluate treatment credibility. Credibility was assessed once, at the middle of Session 1, following the presentation of the overall treatment rationale and before the intervention itself. Patients rated the therapy on a scale from 0 (not at all) to 8 (very), assessing both how they felt about the intervention and how effective they believed it would be. The scale has demonstrated strong internal consistency and test–retest reliability (Devilly & Borkovec, 2000).

Following reviewer feedback, we conducted a manualized content-coding procedure to assess core-threat engagement in the writing exercises. Two independent judges classified each participant's four-wave exercise set as either *engaged* or *not engaged*, using a conservative binary rule: participants were coded as engaged if at least one wave explicitly and emotionally addressed their pre-identified core threat (see supplementary materials). All participants were coded by the first judge, and a random subset of 20 participants (40%) was independently coded by the second judge to evaluate inter-rater reliability. Inter-rater agreement was substantial (Cohen's $\kappa = 0.70$, 95% CI [0.40, 1.00], 85.7% agreement).

Procedure

After screening, participants completed four 20-minute writing sessions spaced approximately two days apart. Follow-up assessments were conducted one week and two months after completion of the final session. The waitlist control group completed questionnaires at time points equivalent to the active conditions' initial session, fourth session, and one-week follow-up.

Anxiety symptoms, measured using the OASIS, were assessed before each writing session, except for the final session, in which symptoms were assessed immediately after writing. Additionally, symptoms were measured at both follow-up sessions. Functional impairment, measured using the Work and Social Adjustment Scale (WSAS), was assessed at pre-intervention, post-intervention, and at both follow-ups. Mechanistic variables were assessed at pre-intervention (prior to the first writing session) and post-intervention (after the fourth writing session).

Before the first writing session, participants identified their primary core threat using the Core Threat Structured Interview (CTSI). The CTSI is a structured and psychometrically validated online assessment designed to elicit the individual's ultimate feared outcome underlying anxiety or avoidance (Zlotnick & Huppert, 2026). It guides participants through examples and clarification prompts to ensure that the selected threat is personally meaningful and specific enough to serve as the focus for the writing exercises. Finally, participants developed a concise script creating a clear narrative of how these threats were expected to unfold from their proximal threats.

Participants were randomized into two groups: WET and NIW, they were later compared with a waitlist group.

Intervention

Intervention Development and Acceptability: The intervention was developed iteratively through repeated pilot administrations (two to three participants per cycle), each followed by structured feedback regarding clarity, difficulty, and engagement. Revisions were made across cycles until the protocol was judged acceptable for implementation. Acceptability and perceived usefulness were assessed in the present study using post-intervention and follow-up self-report items.

Written Exposure Therapy (WET). Participants received brief exposure-based psychoeducation emphasizing that repeated engagement with distressing material reduces emotional impact (cf. Benito et al., 2024). They then repeatedly expanded their personalized threat script through detailed writing focused on feared content, attending to their thoughts and emotions.

Neutral Imagery Writing (NIW). Participants received psychoeducation emphasizing distraction and mental control in emotional regulation and were guided to describe a neutral situation using vivid sensory detail while avoiding emotional or threat-related content.

Waitlist Control. The waitlist group served as an inactive control and completed the OASIS and WSAS at three one-week intervals corresponding to the pre-intervention, post-intervention, and one-week follow-up assessments of the active conditions.

Full instructions for the WET and NIW conditions, including psychoeducation materials, are provided in the supplementary materials.

Statistical Analysis

Change Model: We employed a multilevel ANCOVA design to account for the nested structure of the data, where anxiety is measured repeatedly within individuals (Bodner & Bliese, 2018; van Breukelen, 2013). By accounting for both within- and

between-subject variability, multilevel modeling reduces bias providing more precise estimates of treatment effects. Furthermore, it becomes possible to investigate individual as well as group level effects (Raudenbush & Bryk, 2002).

To determine treatment efficacy we examined the *fixed effects* — the average rates of change across individuals. As described below, to understand covariation between symptoms and processes we examined the *random effects* — the individual deviations from the group-level trends. To account for follow-up measurements at varying time lags (e.g., 1 week, 2 months), we employ a piecewise growth model, allowing for separate slopes at different follow-up times (Bollen & Curran, 2006; **andersson35yearFollowInternetdeliveredCognitive2013?**). Combined with the measurement model described below, this is in essence a latent change score model (Cáncer et al., 2021; McArdle, 2009).

The equation used was:

$$\theta_{ti} = \beta_{0i} + \beta_{1i} \cdot \min(t-1, 3) \cdot 1_{\{t>1\}} + \beta_{2i} \cdot 1_{\{t>5\}} + \beta_{3i} \cdot 1_{\{t=6\}}$$

$$\beta_{ji} = \bar{\beta}_{j,\text{condition}} + u_{ji}$$

The latent symptom severity (θ) was linearly modeled as a baseline effect (β_{0i}) for each individual and the rate of change between the first and fourth sessions (β_{1i}). Separate terms (β_{2i}) and (β_{3i}) represent the rate of change at follow-up (1 week, 2 months). The individual-specific effects, (β_{ji}), are modeled as deviations from the group-level effects, ($\bar{\beta}_{j,\text{condition}}$).

Measurement Model: Given that anxiety was measured using an ordinal scale, we linked latent symptom severity (θ) to the observed ordinal responses using an ordered logit model with a cumulative link function (McCullagh, 1980). This model assumes that ordinal

responses are generated by an underlying continuous latent variable, with thresholds (κ) defining the response categories. The ordered logit function estimates the probability (ϕ) of endorsing each response category based on the latent symptom severity trait (θ).

To account for measurement error and the multidimensionality of the anxiety construct, we adopted a graded response model (GRM) from item response theory (IRT) (Samejima, 2016). The GRM is well-suited for ordinal data, modeling the probability of endorsing an item category as a function of both item-specific parameters (discrimination: (α), and difficulty: (β) and the individual's latent trait level (θ).

The GRM allows us to model the latent symptom severity (θ) for each individual at a specific time point, while accounting for the unique characteristics of each item. The discrimination parameter (α) captures how well each item distinguishes between individuals with different levels of anxiety, while the difficulty parameter (β) reflects the trait level required to endorse higher response categories. This approach helps address known issues with the reliability of change scores (Prieler & Raven, 2008; Rodebaugh et al., 2016).

In this framework, θ is estimated from the multilevel model described earlier, capturing both individual-specific (random) effects and group-level (fixed) effects. The integration of the GRM with the broader multilevel model allows us to model the latent trait accurately, taking into account both item-level variability and individual differences in anxiety over time.

The equation for the measurement model is as follows:

$$R_{itq} \sim \text{Ordered-logit}(\phi_{itq}, \kappa)$$

$$\phi_{itq} = \alpha_q \cdot (\theta_{it} - \beta_q)$$

Here, the likelihood function represents the probability of observing the ordinal

response (R), given the latent symptom severity (θ). The IRT model specifies the relationship between the latent trait and the item parameters. Detailed specifications for this model and the full code is available in the supplementary materials.

Mechanisms: Our second research question investigates the role of mechanisms in change in pathological anxiety. Following Kazdin's (2007) framework for mechanisms research, the present study intentionally employed a broad exploratory approach to identify potential targets for future confirmatory work. While establishing causality is difficult (see Eronen, 2020), our goal is to identify correlations that may signal candidate causal relationships.

Mechanisms of change in psychotherapy are processes or events through which psychological interventions affect symptoms or functioning. Demonstrating that an intervention causes change does not explain *why* this change came about. Establishing that a process is a mechanism requires association between the intervention and process, and between the process and the symptoms (Kazdin, 2007). Thus, we first establish which processes are affected by the intervention. Next, we examine the relationship between the process and the outcome.

To identify processes that may mediate change, we examined both fear-specific and fear-agnostic mechanisms previously linked to exposure-based interventions. Reductions in fear, indexed by state anxiety during imagining the core threat (BSAM; Berg et al., 1998), reflect habituation or extinction processes. Broader regulatory and cognitive mechanisms included distress tolerance (DTS; Simons & Gaher, 2005), reflecting the ability to endure negative emotion, self-compassion (SCS; Raes et al., 2011), representing adaptive self-relating and emotional regulation; and thought–action fusion (TAF-R; Shafran et al., 1996), the belief that thoughts can cause or prevent real outcomes. Metacognitive beliefs (MCQ-30; Wells & Cartwright-Hatton, 2004) and core schemas (BCSS; Fowler et al., 2006) indexed higher-order beliefs about control, self, and others. Finally, imagery-specific

reactions (IFES; Deeprose et al., 2011) captured distress, intrusiveness, and avoidance in response to personally salient imagery. Together, these constructs span both targeted and transdiagnostic mechanisms, allowing a comprehensive view of how WET may promote change across emotional, cognitive, and regulatory domains.

We focus on the process of change, thus we're interested in the association between *changes* in processes and *changes* in symptoms. To examine this association we use multivariate latent curve models (Baldwin et al., 2014; Damian, 2010; MacCallum et al., 1997). This approach extends univariate growth models by allowing the simultaneous examination of multiple outcomes, capturing both the relationships between individual processes and how they evolve over time. Specifically, we model both anxiety (as measured by the OASIS) and the process of interest in parallel, using a single covariance matrix to estimate the random effects for both processes. By incorporating a shared covariance matrix, we capture the correlations between baseline values (intercepts) and changes (slopes) both within and across the two outcomes. This enables us to estimate the correlation between changes controlling for baseline values and accounting for measurement error. Detailed specifications for this model and the full code are available in the supplementary materials.

Although this analysis does not establish causality, it offers valuable insights into potential intervention targets. For example, identifying processes that strongly correlate with changes in anxiety could inform more focused treatments by highlighting which processes to target for clinical improvement. Further research will need to extend these findings by extending them into full mediation models (Hilley & O'Rourke, 2022).

Missing Data and Estimation

All participants who began the intervention were included in the analysis. Missing data were imputed via Bayesian imputation, assuming that missingness may be affected by baseline trait levels and task completion difficulty.

We used Bayesian methods to estimate the parameters of the model using Markov Chain Monte Carlo (MCMC) algorithms implemented in Stan (Carpenter et al., 2017), and via the rethinking interface (McElreath, 2020). We specified weakly informative priors for the regression coefficients, the thresholds, and the item parameters. We assessed the convergence and the fit of the model using posterior predictive checks, trace plots, and effective sample sizes. We additionally assessed measurement-level fit using pointwise WAIC and PSIS: observations with WAIC pointwise penalty greater than 1 *and* PSIS Pareto k greater than .5 were flagged for inspection, with potential exclusion if such observations were few and the model failed to fit them. Across all 27 models in the study, flagged observations were rare (under 1% of all measurements; under 1% in any single model) and dispersed across measures, participants, and items; we therefore retained all observations. Per-model diagnostics are reported in the supplementary material. We use a credible interval of 89% (McElreath, 2020).

Results

Participants

Recruitment began August 6, 2023, and ended September 25, 2023, and the final assessment occurred December 2, 2023. Of 375 individuals initially screened, 89 were randomly assigned at a 2:1 ratio to WET (n = 60, 67.4%) or NIW (n = 29, 32.6%) conditions. The age of participants in the NIW group was significantly lower than that of participants in the WET group (Wilcoxon test, $W = 628.00$, $p = .044$). Fisher exact test indicated no significant group differences in ethnicity or sex, and Wilcoxon tests showed no significant group differences for pre-treatment symptom measures ($ps > 0.36$).

Given the similarity in post-treatment outcomes between the treatment and control groups, we incorporated additional data (n = 73) from a waitlist control group as an idiographic benchmarking group. This data was collected in a similar study conducted between February 1 and February 24, 2023 (Sorka et al., in preparation). This was done in

order to make sure that the changes found during the interventions were not due to chance. These participants were randomly assigned to a waitlist control group for a study that focused on mechanisms of imagery rescripting. Given the identical inclusion/exclusion criteria and the minor time gap between data collection periods, we expect no systematic differences between the groups. The CONSORT flowchart (see Figure 1) provides a detailed overview of the recruitment, randomization, and follow-up processes across the two active groups. Comprehensive demographic information is provided in Table 1.

The large majority of participants (n = 73, 82.0%) completed all assessments. Study completion rates did not differ between treatment (81.7%) and control (82.8%); Fisher's exact test, $p > .999$. Female participants had a significantly lower attrition rate (11.5%) compared to males (32.1%); Fisher's exact test, $p = 0.034$. Fisher's exact and Wilcoxon tests revealed no significant differences between study completers and non-completers in terms of age, ethnicity, or symptom measures ($ps > 0.24$).

Acceptability and Credibility

Both WET and NIW were rated as highly credible on the Devilly Credibility Scale. On the cognitive subscale, WET scored 20.0 (6.5), whereas NIW scored 20.5 (7.1). Similarly, on the feeling subscale, WET scored 8.7 (3.8) and NIW scored 9.4 (4.1).

Most participants reported that the instructions were clear, although a minority indicated difficulty understanding the requirement to return to the same script across sessions. The task was described as difficult by 26% of participants in the WET condition and 25% in the NIW condition, whereas 26% and 42%, respectively, described it as pleasant or useful. At the two-month follow-up, 55% of WET participants and 25% of NIW participants reported improvements in quality of life, and 41% and 38%, respectively, reported continued application of skills learned during the intervention.

Qualitative inspection of the writing materials indicated clear differentiation between

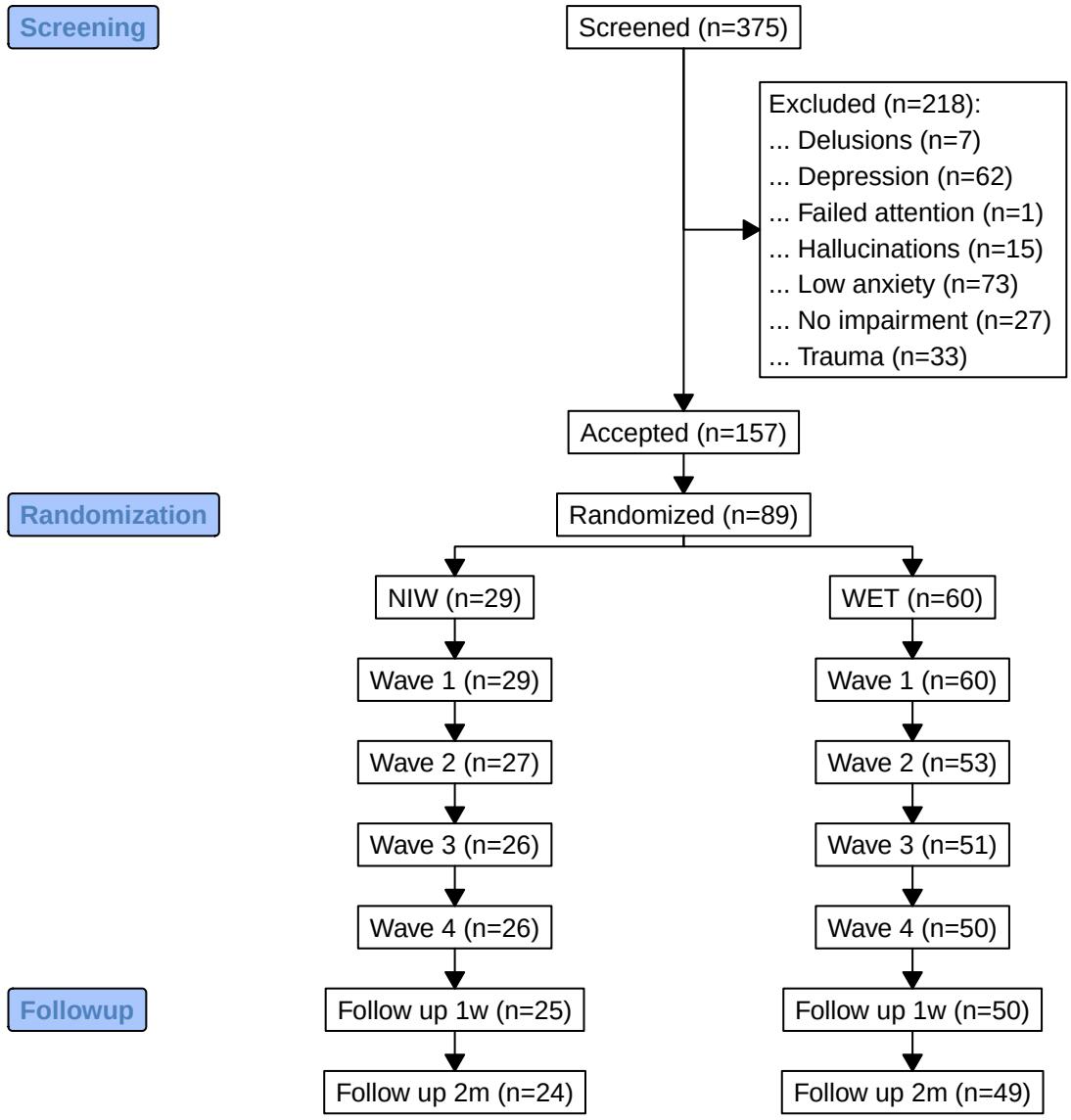

Figure 1

A CONSORT flowchart providing a detailed overview of the recruitment, randomization, and follow-up processes across the two active groups.

Table 1

Pre-treatment and Demographic Characteristics of Sample.

Characteristics	NIW	WET	Waitlist	Total
N	29	60	73	162
Age M (SD)	36 (11.9)	42.6 (15.3)	42.6 (13)	41.4 (13.8)
Gender Female	22 (75.9%)	39 (65%)	48 (65.8%)	109 (67.3%)
Gender Male	7 (24.1%)	21 (35%)	25 (34.2%)	53 (32.7%)
Ethnicity				
Asian	3 (10.3%)	7 (11.7%)	3 (4.1%)	13 (8%)
Black	4 (13.8%)	2 (3.3%)	1 (1.4%)	7 (4.3%)
White	22 (75.9%)	51 (85%)	69 (94.5%)	142 (87.7%)
Country				
Australia	2 (6.9%)	0	0	2 (1.2%)
Canada	2 (6.9%)	0	0	2 (1.2%)
South Africa	4 (13.8%)	0	1 (1.4%)	5 (3.1%)
United Kingdom	20 (69%)	59 (98.3%)	65 (89%)	144 (88.9%)
United States	1 (3.4%)	0	6 (8.2%)	7 (4.3%)
Ireland	0	1 (1.7%)	1 (1.4%)	2 (1.2%)
Employment				
Full-Time	17 (58.6%)	19 (31.7%)	41 (56.2%)	77 (47.5%)
Part-Time	5 (17.2%)	17 (28.3%)	11 (15.1%)	33 (20.4%)
Job Seeking	1 (3.4%)	4 (6.7%)	3 (4.1%)	8 (4.9%)
Not in Workforce	4 (13.8%)	12 (20%)	12 (16.4%)	28 (17.3%)
No Data	2 (6.9%)	8 (13.3%)	6 (8.2%)	16 (9.9%)
Pretreatment measures				
OASIS M (SD)	8.3 (3)	7.6 (2.7)	7.8 (2.8)	7.8 (2.8)
WSAS M (SD)	15.3 (6.8)	12.5 (6.4)	12.5 (6.4)	13 (6.5)

Note. Data presented as number (%) of patients unless otherwise indicated.

OASIS = transdiagnostic anxiety severity; WSAS = functional impairment.

conditions. Participants in the WET condition generally produced emotionally intense scripts directly addressing their core threats. For example, one socially anxious participant wrote:

> What am I doing here? I am a nobody. I know nothing. I feel almost sick now with nerves and dread. What if I just left now? Would anyone even notice? But it's too late... I am to speak first, so I can't even steal other people's ideas... [They're thinking] Who is this idiot? I stumble on and on. I want to stop and for it all to be over, but I can't. It just goes on forever. It is like dying in slow motion.

In contrast, adherence in the NIW condition was less constrained, with some participants producing scripts that emphasized positive outcomes rather than neutral content. For example, one participant wrote:

> I feel enthusiastic about starting my day on a positive and productive day. Today instead of going through my phone for an hour on social media, I start with making my bed and cleaning my room... I start applying for opportunities that align with my qualifications and a bit of me is worried but still hopeful that I get recognized and recruited.

This style of writing resembles techniques used in imagery rehearsal interventions (cf. Krakow & Zadra, 2006).

Symptom Measures

We fit a Bayesian latent growth curve model to assess the efficacy of the interventions (see Figure 2 for a visualization of the model's estimated trajectories). We calculated the standardized mean change (SMC) for each group using the pre-treatment standard deviation as the denominator (Becker, 1988). This standardization allows comparisons across measures, with higher effect size values indicating better outcomes. Additionally, we report the probability of direction, which quantifies the certainty that the outcome is greater than 0.

The probability of direction, marked in this paper as P(>0), ranges from 0 to 1, where values close to 0 indicate high certainty of a negative effect (symptom deterioration), and values close to 1 indicate high certainty of a positive effect (symptom improvement See Makowski et al., 2019).

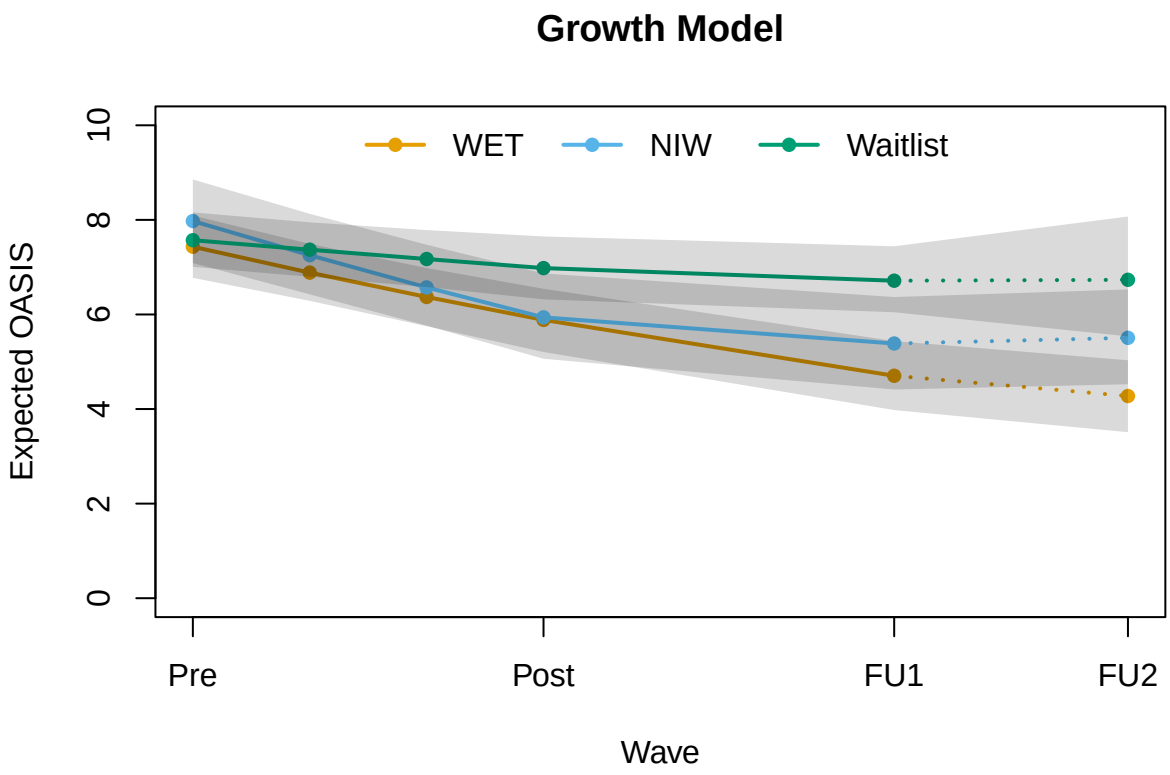

Figure 2

Estimated treatment trajectories for each intervention based on the latent growth model. Values are adjusted for gender. WET = Written Exposure Therapy; NIW = Neutral Imagery Writing; OASIS = Overall Anxiety Severity and Impairment Scale. The time intervals between pre-treatment and post-treatment, and between post-treatment and followup 1 (FU1), were approximately one week. The interval between FU1 and followup 2 (FU2) was approximately two months, which is indicated by the dashed regression line. The shaded gray area represents the 89% credible interval (CI).

Clinical anxiety, as measured by the OASIS, significantly improved across all three

interventions at the one-week follow-up. The estimated raw score improvements were M = 2.73, 89% CI [2.14, 3.35], P(>0) = 100% for the WET group, M = 2.59, 89% CI [1.74, 3.41], P(>0) = 100% for the NIW group, and M = 0.86, 89% CI [0.23, 1.45], P(>0) = 98% for the Waitlist group.

Participants in the WET group showed significant improvements during the treatment period compared to the Waitlist group (d = 0.35, 89% CI [0.09, 0.62], P(>0) = 99%). These gains continued at a similar rate at the one-week follow-up, again compared to the Waitlist group (d = 0.34, 89% CI [0.13, 0.56], P(>0) = 99%). At the two-month follow-up, the gains in WET were estimated to continue at a slower rate compared to the previous gains observed in the WET group (d = 0.16, 89% CI [-0.23, 0.57], P(>0) = 74%). Notably, this rate of change was estimated in relation to the projected Waitlist trajectory, as no Waitlist data was collected at this time point. The improvement trajectory appeared linear, with consistent gains observed throughout the intervention and continuing during both follow-ups. Total improvement when compared to the projected Waitlist trend was d = 0.85, 89% CI [0.37, 1.37], P(>0) = 100%.

Participants in the NIW group also demonstrated significant improvements during the treatment period compared to the Waitlist group (d = 0.47, 89% CI [0.18, 0.76], P(>0) = 100%). These improvements were sustained at the one-week follow-up, again in comparison to the Waitlist group (d = 0.09, 89% CI [-0.14, 0.33], P(>0) = 73%). At the two-month follow-up, the gains were maintained relative to the projected Waitlist trajectory, as no Waitlist data was collected at this time point (d = -0.03, 89% CI [-0.44, 0.38], P(>0) = 43%). Participants in the NIW group showed considerable overall improvement (d = 0.52, 89% CI [-0.02, 1.08], P(>0) = 94%). From pre-treatment to the two-month follow-up, WET demonstrated a slightly greater cumulative advantage over NIW (d = 0.33, 89% CI [-0.05, 0.72], P(>0) = 91%). While participants in the NIW group showed considerable overall improvement (d = 0.52, 89% CI [-0.02, 1.08], P(>0) = 94%), the L-shaped trajectory

suggests that most of the gains occurred during the treatment phase and then stabilized. Contrary to our hypothesis, the NIW group had a slight advantage over WET during the treatment phase (d = -0.11, 89% CI [-0.41, 0.18], P(>0) = 27%). However, WET outperformed NIW at both the one-week follow-up (d = 0.25, 89% CI [0.03, 0.47], P(>0) = 96%) and the two-month follow-up (d = 0.20, 89% CI [-0.09, 0.49], P(>0) = 86%).

These findings suggest that both WET and NIW interventions effectively reduce trait anxiety compared to the Waitlist control. However, while NIW showed slightly greater gains during the treatment phase, WET exhibited stronger and more sustained improvements post-treatment. This pattern highlights the potential of WET for achieving longer-term symptom reduction.

Functional impairment, assessed using the WSAS, showed significant improvement across all three interventions. At one-week follow-up, the estimated improvement in WSAS scores was M = 3.39, 89% CI [2.20, 4.60], P(>0) = 100% for WET, M = 3.53, 89% CI [1.59, 5.36], P(>0) = 100% for NIW, and M = 1.56, 89% CI [0.39, 2.72], P(>0) = 98% for the waiting list. Both WET (d = 0.29, 89% CI [0.03, 0.56], P(>0) = 96%) and NIW (d = 0.27, 89% CI [-0.06, 0.59], P(>0) = 91%) resulted in greater improvements in functional impairment compared to the waiting list at the one week followup. However, no significant differences were observed between them (d = 0.02, 89% CI [-0.31, 0.35], P(>0) = 54%). These outcomes were kept at the 2 month followup.

Engagement

Most (n = 33, 66.0%) of the participants who completed WET were classified as having engaged with their identified core threat at least once across the four writing sessions. Across all stages, participants who engaged showed better outcomes than those who did not.

During treatment, engaged participants showed a small advantage over non-engaged participants (d = 0.32, 89% CI [-0.16, 0.79], P(>0) = 86%). At the one-week follow-up the

difference was larger and statistically reliable (d = 0.37, 89% CI [-0.16, 0.86], P(>0) = 88%), whereas at the two-month follow-up the estimated difference was smaller and less certain (d = 0.29, 89% CI [-0.26, 0.84], P(>0) = 80%).

When compared with the NIW group, outcomes during treatment were similar (d = 0.01, 89% CI [-0.38, 0.40], P(>0) = 51%). A modest advantage was observed for engagers at the one-week follow-up (d = 0.24, 89% CI [-0.18, 0.67], P(>0) = 81%), and by the two-month follow-up the difference was larger and statistically reliable (d = 0.40, 89% CI [-0.05, 0.86], P(>0) = 92%).

Treatment Processes

Our second, exploratory hypothesis proposed that the identified mechanisms of change would improve during therapy. Detailed results for WET are presented in Table 2. Below the table, we highlight the significant findings in WET, alongside the corresponding effects observed in the NIW group. Difference between the groups is reported in all places where they are significant. This comparison aims to identify differential effects while maintaining a primary focus on WET, which is the central topic of this paper. Full results for NIW are available in the supplementary materials.

Distress tolerance (DTS) improved significantly in the WET group (d = 0.29, 89% CI [0.07, 0.51], P(>0) = 98%), while it somewhat deteriorated in the NIW group (d = -0.17, 89% CI [-0.41, 0.08], P(>0) = 14%). The difference between the groups was statistically significant (d = 0.46, 89% CI [0.14, 0.75], P(>0) = 99%), aligning with our expectations. WET likely encouraged participants to tolerate distress through structured exposure, whereas NIW may have reinforced distress tolerance and avoidance.

Positive self-schemas, as measured by the BCSS, increased significantly in WET (d = 0.23, 89% CI [0.09, 0.37], P(>0) = 99%), suggesting that the intervention fostered more adaptive self-beliefs. In contrast, no significant change was found in the NIW group (d =

Table 2

Process estimates for each of the proposed mechanisms in WET.

	Mean		Change		
Measure	Pre	Post	*d*	89% CI	P(>0)
DTS †	24.43	27.71	-0.29	[-0.51,-0.07]	0.02*
SCS †	20.45	20.43	0.00	[-0.13,0.14]	0.51
IFES	29.99	25.96	0.31	[0.13,0.49]	1.00*
TAF					
Morality	15.07	15.05	0.00	[-0.20,0.21]	0.52
Likelihood	3.79	2.81	0.14	[-0.03,0.31]	0.90
BCSS					
Self negative	4.25	3.80	0.11	[-0.05,0.26]	0.86
Other negative	6.10	6.12	-0.01	[-0.21,0.19]	0.48
Self positive	11.13	12.39	-0.23	[-0.37,-0.09]	0.01*
Other positive	13.07	13.40	-0.08	[-0.24,0.09]	0.22
MCQ					
(Lack of) Cognitive Confidence	3.43	2.21	0.29	[0.12,0.47]	0.99*
Positive Beliefs about Worry	4.10	4.09	0.00	[-0.18,0.19]	0.50
Cognitive Self-Consciousness	8.67	8.55	0.03	[-0.13,0.19]	0.62
Uncontrollability and Danger	6.62	5.71	0.21	[0.08,0.35]	1.00*
Need to Control Thoughts	4.58	4.29	0.09	[-0.08,0.26]	0.79

Note. DTS = Distress Tolerance; SCS = Self-Compassion; IFES = Impact of Future Events; TAF = Thought–Action Fusion; BCSS = Beliefs about Self and Others; MCQ = Metacognitive Beliefs. Higher scores indicate greater symptom severity, except for measures marked with †, which are strength-based.

0.11, 89% CI [-0.08, 0.30], P(>0) = 84%). Perceived impact of future events (IFES) decreased significantly in the WET group (d = 0.31, 89% CI [0.13, 0.49], P(>0) = 100%). No significant change was observed in the NIW group (d = 0.05, 89% CI [-0.14, 0.25], P(>0) = 67%). No significant between group differences were found for both of these.

For metacognitions measured by the MCQ, both maladaptive indices-(lack of) cognitive confidence (d = 0.29, 89% CI [0.12, 0.47], P(>0) = 99%) and negative beliefs about uncontrollability and danger (d = 0.21, 89% CI [0.08, 0.35], P(>0) = 100%) increased significantly in WET. However, no significant changes were found in the NIW group for either (lack of) cognitive confidence (d = -0.08, 89% CI [-0.34, 0.16], P(>0) = 30%) or negative beliefs about uncontrollability and danger (d = 0.07, 89% CI [-0.08, 0.22], P(>0) = 77%). The only significant between group difference was found in (lack of) cognitive confidence (d = -0.38, 89% CI [-0.69, -0.10], P(>0) = 2%).

These findings suggest several potential mechanisms of change for WET, which we investigate in the upcoming section. The other mechanisms did not improve and thus were not analyzed.

Mechanisms of Change

We examined each process affected by WET to assess whether changes in these processes corresponded with symptom change. Detailed results, including specific changes in processes and symptom levels, can be found in Table 3.

The only process found both to respond to WET and to change alongside symptoms was distress tolerance (DTS). The correlation between changes in DTS and symptoms during treatment was r = -0.35, 89% CI [-0.61, -0.06], P(>0) = 3%. Given that NIW actively *decreased* distress tolerance, we investigated the effect of change in distress tolerance on change in symptoms within NIW as well. We found that in contrast with WET, in NIW an increase in distress tolerance was actually correlated with an increase in symptoms r = 0.29,

Table 3

Proposed mechanisms of change. Correlations between symptom change (during treatment and follow-up) and processes that have been found to change in WET.

	Treatment			Followup		
Measure	*r*	89% CI	P(<0)	*r*	89% CI	P(>0)
DTS †	-0.37	[-0.39, -0.34]	0.03*	0.13	[0.11, 0.16]	0.74
BCSS Self positive	-0.23	[-0.25, -0.20]	0.15	-0.31	[-0.34, -0.27]	0.10
IFES	0.26	[0.23, 0.29]	0.92	-0.07	[-0.10, -0.04]	0.37
(Lack of) Cognitive Confidence	0.13	[0.10, 0.16]	0.72	-0.29	[-0.32, -0.26]	0.13
Uncontrollability and Danger	0.42	[0.39, 0.46]	0.91	-0.45	[-0.48, -0.41]	0.07

Note. DTS = Distress Tolerance; IFES = Impact of Future Events; BCSS = Beliefs about Self and others. Higher scores indicate greater symptom severity, except for measures marked with †, which are strength-based.

89% CI [-0.03, 0.59], P(>0) = 93%. This suggests a differential process where in WET distress tolerance decreases anxiety whereas in NIW anxiety is decreased via sensitization to distress, likely via distraction and avoidance.

No processes were significantly predictive of change at follow-up. However, improvements in positive self-schema (r = -0.30, 89% CI [-0.63, 0.07], P(>0) = 10%), cognitive confidence (r = -0.28, 89% CI [-0.64, 0.11], P(>0) = 13%), and a reduction in negative beliefs about uncontrollability and danger (r = -0.42, 89% CI [-0.78, 0.06], P(>0) = 7%) all weakly predicted change at follow-up.

Thus, these four processes warrant additional research when investigating the mechanisms of WET.

Response to Core Threats

Alongside global mechanisms and metacognitions, we explored how participants' threat responses changed in relation to their core threat. To investigate this, we examined state anxiety before and after each writing session using the BSAM. Participants rated their state anxiety while vividly imagining their core threat script. This approach allowed us to assess both within-session and between-session changes in anxiety levels (see Figure 3).

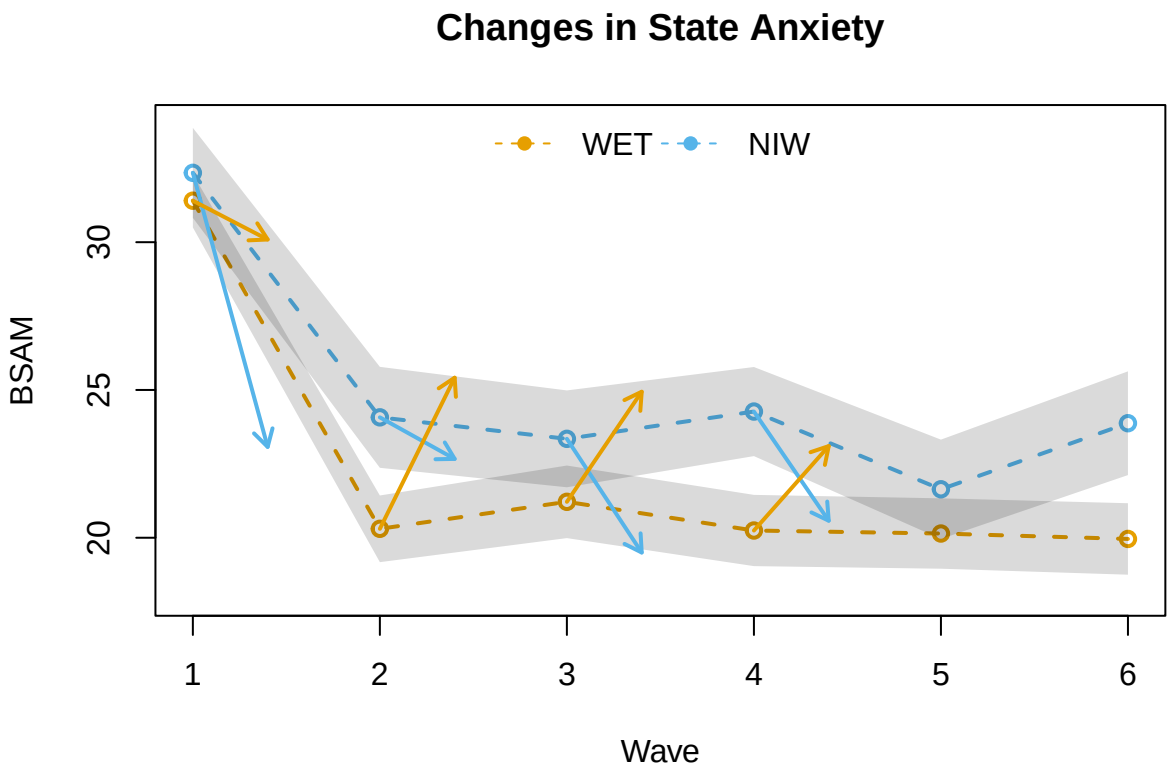

Figure 3

Trajectory of state anxiety as a function of time. Dashed lines are the trajectory of pre intervention measurements. Arrows are within session changes, pointing at the post intervention measure.

Our findings revealed distinct differences between the groups. Both began with similarly high levels of state anxiety. During the first session, the NIW group experienced a

significant reduction in anxiety, reflecting their instruction to distract themselves from the threat. In contrast, the WET group's anxiety levels remained high and stable, consistent with their instruction to focus on the core threat.

By the second session, both groups had established new patterns: the NIW group maintained the reduced anxiety achieved after session one, while the WET group experienced a significant reduction in anxiety even below that of the NIW group. This between-session reduction in state anxiety predicted improvements in trait anxiety (OASIS), in line with meta-analysis showing that between session fear reduction is correlated with treatment outcome (e.g., Rupp et al., 2017). However, stronger correlations were found in the WET group (r = 0.39, 89% CI [0.13, 0.61], P(>0) = 99%) than in the NIW group (r = 0.28, 89% CI [-0.04, 0.58], P(>0) = 92%). It is reasonable to hypothesize that the reduction in state anxiety in response to the core threat is caused by decrease in DTS, as seen above. However, no correlation was found between change in DTS and change in state anxiety (r = -0.03, 89% CI [-0.32, 0.26], P(>0) = 43%).

The distinct trajectory of state anxiety in the WET group suggests a unique mechanism of change. Specifically, the larger decrease in state anxiety between sessions in the WET group may indicate deeper processing of core threats, which in turn is associated with more robust symptom improvements.

Furthermore, within-session patterns diverged between the groups. For the NIW group, writing reduced state anxiety, while for the WET group, it increased state anxiety. This result serves as a manipulation check and aligns with the interventions' instructions: WET participants were encouraged to actively engage with their fears, while NIW participants were guided to distract themselves from them.

Interestingly, there was evidence of a slight reduction in the magnitude of within-session anxiety increases over time in the WET group. This trend suggests that

participants may have become more accustomed to the task, potentially reflecting a process of habituation or extinction.

Acceptability and Engagement

In the present study, most participants reported that the instructions were clear; a minority indicated difficulty understanding the instructions to return to the same script. Task difficulty was reported by 25% of participants in the NIW condition and 26% in the WET condition, whereas 42% and 26%, respectively, described the task as pleasant or useful. At the two-month follow-up, 25% of NIW participants and 55% of WET participants reported improvements in quality of life, and 38% and 41%, respectively, reported continued application of skills learned during the intervention.

A trained judge rated whether participants meaningfully engaged with their identified core threat at any point across the four writing sessions. Thirty-one of fifty participants were judged to have engaged with their core threat. Participants who engaged tended to show greater improvement, particularly at the two-month follow-up.

Discussion

This study suggests that self-guided, future-oriented WET is an effective and scalable intervention for treating transdiagnostic anxiety symptoms. Moreover, it highlights distress tolerance and anxiety levels regarding one's core threat as key mechanisms of change.

Effectiveness of WET

Recent years have seen a growing interest in evidence-based self-help interventions, including unguided (e.g., Morgan et al., 2017) and guided online treatments (Andersson et al., 2019). These interventions often achieve effect sizes comparable to face-to-face therapy (Hedman-Lagerlöf et al., 2023). Our findings contribute to this literature by demonstrating that self-guided, online, future-oriented WET can achieve meaningful improvements. This raises potential for the incorporation of WET in transdiagnostic or specific online protocols.

In a meta-analysis of online unguided self-help interventions, Wang and colleagues

(2023) reported small effects when treating anxiety ($d = 0.35$, 95% CI [0.22, 0.48]). Whereas in the current study, WET demonstrated a small effect size compared to waitlist control during treatment ($d = 0.35$) it developed into a medium effect size at one week follow-up ($d = 0.69$) and kept improving at the two month followup ($d = 0.85$). This change corresponds to approximately 3.16 points on the OASIS, which is more than half the reliable change threshold of 4.41 (Moore et al., 2015). Alongside the reduction in anxiety, we observed a small but significant improvement in functional impairment, corroborating the clinical relevance of the intervention. WET produced clinically meaningful symptom reduction, but the magnitude of change fell short of full reliable change for many participants and was accompanied by only modest functional improvement. Accordingly, WET may be best positioned as a first-line or adjunctive intervention within a stepped or multi-component treatment approach rather than as a stand-alone intervention.

Previous research has established WET's efficacy for specific disorders. For example, Sloan and Marx demonstrated non-inferiority to best-practice treatments for PTSD (Sloan et al., 2023; Thompson-Hollands et al., 2019). Similarly, future-oriented WET has shown promise for treating generalized anxiety disorder (Goldman et al., 2007; Ovanessian et al., 2019). Our findings extend these principles to anxiety as a transdiagnostic phenomenon. Moreover, consistent with previous studies (Sloan et al., 2018; Thompson-Hollands et al., 2018), we observed continued improvements post-intervention, highlighting WET's potential for sustained effects. The diversity of our sample and standardized online delivery suggest these findings should generalize well to other populations. However, our unusually low dropout rates may be influenced by participant compensation, and cultural differences or technology limitations could affect the intervention's effectiveness in other contexts.

The effectiveness of WET supports the notion that writing about one's future threats enables emotional processing. Furthermore, it highlights that individuals are capable of engaging in this process without the active guidance of a therapist. Beyond the implications

as an inexpensive and accessible intervention, WET provides a scalable method to investigate therapeutic processes in exposure and other interventions. It offers a relatively simple intervention that can easily be adjusted to emphasize different aspects of the intervention process, such as focusing on the first or third person, varying levels of immersion, or addressing past versus future events. Additionally, WET generates rich textual data that can offer further insights into the cognitive and emotional dynamics at play during emotional processing Thus, WET serves as an important tool in the psychotherapy researcher's toolbox.

Beyond symptom change, both interventions were rated as highly credible, suggesting that participants perceived them as logical and emotionally compelling. Although a minority of participants described the writing task as difficult, overall acceptability was high, and many participants—particularly in the WET condition—reported improvements in quality of life and continued application of learned skills at follow-up.

Exploratory subgroup analyses further suggested that participants who engaged directly with their identified core threat during the writing exercises showed greater short-term improvement, with reliable advantages at one-week follow-up and smaller, less certain differences thereafter. Although consistent with mechanistic accounts of WET, these findings should be interpreted cautiously given the small subgroup and post-hoc nature of the coding, and therefore remain preliminary. Future work should evaluate engagement prospectively and test whether encouraging such engagement can strengthen clinical response. Together, these findings suggest that self-guided WET is not only effective but also acceptable and feasible, while underscoring the importance of optimizing instructional clarity and engagement to maximize longer-term benefit.

Notes on Technique

The WET protocol differs significantly from traditional, future-oriented imaginal exposure flooding methods, introducing both key innovations and trade-offs (see Asnaani et al., 2016). Traditional imaginal exposure typically involves recording a fixed threat script

that patients listen to repeatedly fostering immersion by encouraging patients to close their eyes and focus inward. In contrast, the WET protocol requires participants to rewrite their threat script each session. This approach creates an evolving narrative that may facilitate processing of the meanings associated with the threat (see Sloan & Marx, 2019).

This approach introduces a tradeoff between immersion and reflective distance. While traditional imaginal exposure enhances immersion through closing ones' eyes, WET requires writing with ones' eyes open, which may reduce immersion but adds a layer of reflective distance. To address this, participants were instructed to remain attuned to their emotions and sensations, and if they run out of content, briefly close their eyes to regain immersion.

A unique challenge in imaginal exposure is the difficulty in assessing participants' levels of engagement and immersion. WET addresses this issue by creating a concrete, written record of participants' thoughts and experiences during exposure. Clinically, this written documentation provides valuable feedback, allowing therapists to guide participants on specific aspects to enhance (for example, advising them to focus more on physical sensations if needed). The written records are also useful for research, enabling analysis of factors like exposure duration, narrative detail, and the specific language used to describe the threat.

NIW as an Intervention

WET did not appear to improve immediate outcomes relative to the control condition. However, we believe this interpretation misses an important point: NIW showed significant effects, suggesting it may not have been an inert placebo. Active control conditions in psychological interventions typically demonstrate moderate effects. For example, research on internet-based interventions for anxiety has shown that active control yield effect sizes around 0.5, using interventions such as relaxation tasks or forum participation (Carlbring et al., 2011; Johansson et al., 2012; Newby et al., 2018). By contrast, NIW had an absolute effect size of 0.86 at the one week follow-up. This suggests

that NIW might not simply be a placebo.

Several mechanisms may explain NIW's efficacy. One possibility is the role of *distancing*, a well-documented emotion regulation strategy (Kross & Ayduk, 2017; Powers & LaBar, 2019). Writing about non-distressing topics may have helped participants practice disengaging from threatening cognitions, facilitating emotional recovery. Indeed, such training reduces distress (Denny & Ochsner, 2014), and this was the faux rationale given in the psychoeducation section. Alternatively, having participants evaluate their response to the core threat script twice per session may have served as a form of exposure. Additionally, NIW scripts often included positive or meditative content, suggesting that mindfulness or positive imagery may have supported the observed benefits (Hofmann et al., 2010; Pile et al., 2021). Similarly, rehearsing positive ideation, even when unrelated to personal worries, has been shown to counteract worry in individuals with generalized anxiety disorder (Eagleson et al., 2016).

Baseline anxiety levels may have also influenced NIW's effectiveness. NIW participants began with higher anxiety scores, potentially amplifying improvements. Greater baseline severity often yields larger gains due to regression to the mean (Barnett et al., 2005) or more accessible pathology (Driessen et al., 2010). This difference remains a potential confounder despite lacking statistical significance.

NIW shows promise as a therapeutic tool for training distancing, distraction, and disengagement skills. However, we found contrasting trajectories with WET: NIW produced rapid initial anxiety reduction but plateaued over time, while WET showed slower initial progress but continued steady symptom reduction, ultimately achieving superior long-term effectiveness. This suggests that NIW may offer immediate symptom relief, while WET provides more benefit over time. The small sample size in the NIW condition (29 participants) limits generalizability, emphasizing the need for further research to confirm these effects and determine optimal contexts for each approach.

How WET Works

The current study aims to advance our understanding of treatment mechanisms by identifying potential mediators. Mediators are potential pathways linking treatment to outcomes, though not all mediators qualify as true causal mechanisms (Kraemer et al., 2002). We surveyed multiple potential processes to identify candidates for change in WET. Among those that significantly changed—distress tolerance, positive self-schema, perceived impact of future events, and metacognitive processes—only distress tolerance was significantly correlated with symptom reduction, suggesting it as a key mechanism. Additionally, change in the fear response to core threats between the first and second session predicted symptom change.

Distress Tolerance

Distress tolerance, defined as an individual's ability to endure negative emotional states (Simons & Gaher, 2005), is increasingly recognized as a transdiagnostic factor of emotional disorders. It has been linked to symptom severity and treatment outcome in conditions including depression (Lass & Winer, 2020), anxiety (Michel et al., 2016), PTSD (Boffa et al., 2018), and substance use disorders (Leyro et al., 2010). Psychotherapies that enhance distress tolerance, such as dialectical behavior therapy, often improve patients' capacity to engage with distressing emotions, reducing avoidance behaviors and promoting resilience (Leyro et al., 2010). Indeed, prior research has demonstrated that distress tolerance mediates change in mindfulness-based interventions (Li et al., 2023, 2024).

Our findings suggest that WET fosters distress tolerance presumably by encouraging participants to engage with distressing thoughts and emotions. While these results support the role of distress tolerance as a mechanism of change in WET, further research is needed to establish a causal connection between changes in DTS and symptom reduction (Kraemer et al., 2002). Specifically, longitudinal studies could clarify whether improvements in distress tolerance precede or occur concurrently with symptom changes.

Research suggests specific distress tolerance subtypes (intolerance of uncertainty, anxiety sensitivity) predict change better than general DTS (Michel et al., 2016). This necessitates refining our understanding of how WET influences distress tolerance and its role in symptom reduction. Future research should explore these subdomains to clarify how WET enhances specific facets of distress tolerance and their contribution to therapeutic outcomes. A nuanced approach may reveal distinct change pathways, informing tailored interventions that maximize impact across emotional disorders.

Responses to Core Threats

Central to pathological fear is an inflated sense of threat, making fear reduction key to alleviating anxiety (Zlotnick & Huppert, 2025). Our findings showed WET reduced fear responses to core threats more effectively than NIW. The rate of fear reduction between Sessions 1 and 2 predicted symptom improvement by Session 4, supporting temporal precedence.

Most change in fear responses occurred early, with minimal additional reduction thereafter despite continued symptom improvements. This pattern suggests that early engagement with threat material may rapidly reduce fear, while later sessions might improve these gains through processes such as distress tolerance or mastery. Future research should test the dose–response relationship between intervention exposure and the degree of core threat fear reduction, and determine whether such early mechanism change is sufficient to produce enduring symptom improvement. Additionally, future iterations could examine whether refining writing scripts to better target "hot spots"—as done in WET for PTSD (Sloan & Marx, 2019)—enhances the maintenance or magnitude of threat-reduction effects.

Participant engagement predicted improvement particularly at the two month follow-up. Although this analysis was exploratory and not preregistered, it suggests that enhancing engagement—for example, through adaptive or self-feedback mechanisms—may strengthen treatment effects. Future work will evaluate inter-rater reliability and imagery

immersion (e.g., present-tense, sensory, and emotional content) as complementary markers of engagement quality.

Emotional engagement itself may represent a key driver of therapeutic change Focusing on emotional rather than cognitive aspects of writing enhances outcomes in PTSD (Sloan et al., 2007). Future iterations of WET could deepen emotional processing at each session by explicitly encouraging emotional engagement or enhancing imagery realism (Holmes & Mathews, 2005, 2010). Furthermore, improving the threat script by better reflecting the core threat and decreasing avoidance may also be important (Zlotnick & Huppert, 2025).

The distinct threat-response trajectories in WET and NIW illustrate their divergent mechanisms. In WET, sessions elevated threat-response, but the between-session reduction led to lower overall threat-response. In contrast, NIW provided short-term threat-response relief with smaller long-term benefits, likely reflecting its strategy of distraction rather than engagement with the core threat. This highlights the importance of targeting core threats in interventions designed to reduce anxiety.

Changes in both threat response and distress tolerance predicted symptom reduction. One might hypothesize these processes are related—perhaps decreased threat response results from increased distress tolerance. However, we found no significant correlation between these variables, suggesting they operate independently. This distinction highlights two complementary WET mechanisms: distress tolerance as a generic skill applicable across distinct threats, and threat response changes that address specific threats manifesting across various contexts. Future research should investigate how these processes interact and complement each other to influence treatment outcomes, potentially providing broader insights into therapeutic mechanisms beyond WET.

Several limitations should be noted. The waitlist group was drawn from a prior study

rather than randomized concurrently, which may introduce time-related bias despite identical criteria. Although mechanisms were preregistered, the analyses do not establish causality, and the number of exploratory tests increases the risk of false positives. All measures were self-reported online, limiting external verification. Participants were paid volunteers recruited through an online platform and likely less symptomatic and more technologically oriented than typical clinical populations, limiting generalizability to more severe or high-risk groups where assessment of suicidality and clinical management are essential. These limitations are balanced by preregistration, consistent methodology across cohorts, and replication of key effects in independent samples.

Finally, the intervention targeted a single core threat identified by each participant. We believe this approach captures the central motivational theme underlying multiple situational fears and thus provides efficient coverage of the broader threat system (Zlotnick & Huppert, 2025). Nevertheless, a single-target design cannot encompass all personally relevant threats, and future studies should examine whether addressing multiple core threats sequentially or hierarchically yields added benefit.

Advancing Mechanism-Driven Research with Transdiagnostic WET

This study demonstrated self-guided, online transdiagnostic WET's efficacy, showing superiority over both waitlist and active writing controls. With just four 20-minute writing sessions, WET achieved large effect sizes at follow-up. The study identified two potential mechanisms: increased distress tolerance and reduced distress responses to core threats.

As clinical psychology shifts from outcome-focused to hypothesis-driven mechanism research (Hofmann & Hayes, 2019), flexible methodologies are increasingly needed. Transdiagnostic WET meets this need as a concise, self-administered intervention easily delivered online. Beyond its effectiveness in reducing anxiety and functional impairment, WET's adaptability is its strength. Its structured yet flexible instructions can be tailored to test various interventions, target different mechanisms, and assess effects across outcomes.

WET enables investigation of both processes and outcomes across diverse populations and settings, advancing mechanism-driven research while offering a scalable, accessible transdiagnostic intervention.

Author Contributions

- **Elad Zlotnick:** Conceptualization, Methodology, Data collection, Formal analysis, Writing - Original Draft Preparation, Writing - Review & Editing
- **Hila Sorka:** Conceptualization, Data collection, Writing - Review & Editing
- **Snir Barzilay:** Conceptualization, Writing - Review & Editing
- **Jonathan D. Huppert:** Conceptualization, Supervision, Resources, Writing - Review & Editing, Funding acquisition

Conflicts of Interest

The authors declare that there were no conflicts of interest with respect to the authorship or the publication of this article.

Funding

This work was supported by the Israel Science Foundation (ISF 1905/20) awarded to Jonathan Huppert, Sam and Helen Beber Chair of Clinical Psychology, and by the Society for Psychotherapy Research (SPR) small research grant 2023 awarded to Elad Zlotnick.

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